THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	A point is inside if its distance from O is less than the radius.
2	**(c)** 180	Triangle angle sum.
3	**(c)** 130	Linear pair.
4	**(b)** 5	Outer radius minus inner radius.
5	**(b)** 8	Side = perimeter ÷ 4.
6	**(b)** 24	4 diameters.
7	**(a)** 7	Half of AB.
8	**(c)** 3	A point is inside if its distance from O is less than the radius.
9	**(a)** 34	Opposite sides are equal.
10	**(d)** 110	Linear pair.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Width $= 9 - 4 = 5$ cm (1); Inner diameter $= 8$ cm; outer diameter $= 18$ cm (1).	1+1
12	$\angle AOD = 180 - 40 = 140^\circ$ (1); $\angle OAB = (180 - 40) \div 2 = 70^\circ$ (1).	1+1
13	Rectangle $= 70$ cm^2; Square $= 64$ cm^2 (1); Larger: rectangle (1).	1+1
14	Width $= 12 - 5 = 7$ cm (1); Inner diameter $= 10$ cm; outer diameter $= 24$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of square $= 36$ cm^2 (1); One triangle is one quarter: 9 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
16	Between neighbours: 6 cm (1); First to third: 12 cm (1); Total length $= 6 \times 3 = 18$ cm (1).	1+1+1
17	$l + b = 20$, so $b = 6$ cm (1); It is not a square (2).	1+2
	OR: Area of square $= 64$ cm^2 (1); One triangle is one quarter: 16 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
18	Between neighbours: 8 cm (1); First to third: 16 cm (1); Total length $= 6 \times 4 = 24$ cm (1).	1+1+1
	OR: $l + b = 30$, so $b = 15$ cm (1); It is a square (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 4 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 4 cm and join DC (1); Half diagonal OA $= 3$ cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width $= 8 - 5 = 3$ m (1); Outer diameter $= 16$ m (1); Inner diameter $= 10$ m (1); Check with the compass-drawn figure (2).	1+1+1+2
20	Draw AB = 5 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 5 cm and join DC (1); Half diagonal OA $= 4$ cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width $= 10 - 6 = 4$ m (1); Outer diameter $= 20$ m (1); Inner diameter $= 12$ m (1); Check with the compass-drawn figure (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 6	Outer radius minus inner radius.
2	**(d)** 11	Side = perimeter ÷ 4.
3	**(b)** 32	4 diameters.
4	**(b)** 9	Half of AB.
5	**(b)** 2	A point is inside if its distance from O is less than the radius.
6	**(d)** 42	Opposite sides are equal.
7	**(c)** 70	Linear pair.
8	**(b)** 6	Outer radius minus inner radius.
9	**(a)** 13	Side = perimeter ÷ 4.
10	**(b)** 40	4 diameters.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle AOD = 180 - 70 = 110^\circ$ (1); $\angle OAB = (180 - 70) \div 2 = 55^\circ$ (1).	1+1
12	Rectangle = 60 cm^2; Square = 64 cm^2 (1); Larger: square (1).	1+1
13	Width = $10 - 6 = 4$ cm (1); Inner diameter = 12 cm; outer diameter = 20 cm (1).	1+1
14	$\angle AOD = 180 - 80 = 100^\circ$ (1); $\angle OAB = (180 - 80) \div 2 = 50^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of square = 100 cm^2 (1); One triangle is one quarter: 25 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
16	Between neighbours: 10 cm (1); First to third: 20 cm (1); Total length = $6 \times 5 = 30$ cm (1).	1+1+1
17	$l + b = 28$, so $b = 8$ cm (1); It is not a square (2).	1+2
	OR: Area of square = 144 cm^2 (1); One triangle is one quarter: 36 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
18	Between neighbours: 12 cm (1); First to third: 24 cm (1); Total length = $6 \times 6 = 36$ cm (1).	1+1+1
	OR: $l + b = 36$, so $b = 18$ cm (1); It is a square (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 6 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 6 cm and join DC (1); Half diagonal OA = 5 cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width = $9 - 4 = 5$ m (1); Outer diameter = 18 m (1); Inner diameter = 8 m (1); Check with the compass-drawn figure (2).	1+1+1+2
20	Draw AB = 7 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 7 cm and join DC (1); Half diagonal OA = 6 cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width = $12 - 7 = 5$ m (1); Outer diameter = 24 m (1); Inner diameter = 14 m (1); Check with the compass-drawn figure (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 11	Half of AB.
2	**(b)** 2	A point is inside if its distance from O is less than the radius.
3	**(d)** 52	Opposite sides are equal.
4	**(d)** 50	Linear pair.
5	**(b)** 9	Outer radius minus inner radius.
6	**(c)** 15	Side = perimeter ÷ 4.
7	**(d)** 48	4 diameters.
8	**(b)** 15	Half of AB.
9	**(c)** 3	A point is inside if its distance from O is less than the radius.
10	**(d)** 66	Opposite sides are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rectangle = 72 cm^2; Square = 81 cm^2 (1); Larger: square (1).	1+1
12	Width = $15 - 7 = 8$ cm (1); Inner diameter = 14 cm; outer diameter = 30 cm (1).	1+1
13	$\angle AOD = 180 - 110 = 70^\circ$ (1); $\angle OAB = (180 - 110) \div 2 = 35^\circ$ (1).	1+1
14	Rectangle = 90 cm^2; Square = 100 cm^2 (1); Larger: square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of square = 196 cm^2 (1); One triangle is one quarter: 49 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
16	Between neighbours: 14 cm (1); First to third: 28 cm (1); Total length = $6 \times 7 = 42$ cm (1).	1+1+1
17	$l + b = 31$, so $b = 9$ cm (1); It is not a square (2).	1+2
	OR: Area of square = 256 cm^2 (1); One triangle is one quarter: 64 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
18	Between neighbours: 16 cm (1); First to third: 32 cm (1); Total length = $6 \times 8 = 48$ cm (1).	1+1+1
	OR: $l + b = 42$, so $b = 12$ cm (1); It is not a square (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 8 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 8 cm and join DC (1); Half diagonal OA = 7 cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width = $8 - 3 = 5$ m (1); Outer diameter = 16 m (1); Inner diameter = 6 m (1); Check with the compass-drawn figure (2).	1+1+1+2
20	Draw AB = 9 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 9 cm and join DC (1); Half diagonal OA = 8 cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: Width = $14 - 8 = 6$ m (1); Outer diameter = 28 m (1); Inner diameter = 16 m (1); Check with the compass-drawn figure (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

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